1. Introduction to ROS2

1. Overview of ROS2

ROS2 is the second-generation Robot Operating System, an upgraded version of ROS1 that solves some of the problems in ROS1. The earliest version of ROS2, Arden, appeared in 2017. With continuous version iteration, updates and optimizations, there is now a stable version. Just like ROS1, the choice of the Linux version is also related to the ROS2 version. The corresponding versions are as follows.

ROS2版本	发布时间	维护截止时间	Ubuntu版本
Ardent Apalone	2017.12	2018.12	Ubuntu 16.04（Xenial Xerus）
Bouncy Bolson	2018.7	2019.7	Ubuntu 16.04（Xenial Xerus）、Ubuntu 18.04 (Bionic Beaver)
Crystal Clemmys	2018.12	2019.12	Ubuntu 18.04 (Bionic Beaver)
Dashing Diademata	2019.5	2021.5	Ubuntu 18.04 (Bionic Beaver)
Eloquent Elusor	2019.11	2020.11	Ubuntu 18.04 (Bionic Beaver)
Foxy Fitzroy	2020.6	2023.5	Ubuntu 20.04 (Focal Fossa)
Galactic Geochelone	2021.5	2022.11	Ubuntu 20.04 (Focal Fossa)
Humble Hawksbill	2022.5	2027.5	Ubuntu 22.04 (Jammy Jellyfish)
Iron Irwini	2023.5	2024.11	Ubuntu 22.04 (Jammy Jellyfish)
Jazzy Jalisco	2024.5	2029.5	Ubuntu 24.04（Noble Numbat）

- **Note**: Choose the ROS2 version that matches your Linux distribution. This series of courses targets **Ubuntu 24.04 + ROS2 Jazzy**, and the commands and cases in the course are adapted to the Jazzy environment. Other ROS2 versions may differ in package names, interfaces, and launch files, so they cannot be copied directly.

2. Features of ROS2

2.1. **ROS2 fully supports three platforms**

- Ubuntu
- Mac OS X
- Windows 10

2.2. Implements a distributed architecture

The central master node is removed, enabling distributed discovery of nodes, publish/subscribe, and request/response communication.

2.3. **Supports real time**

2.4. Uses a new version of the programming language

- C++17
- Python 3.12

2.5. **Uses the new build system Ament (ROS uses Catkin)**

2.6. **ROS1 can communicate with ROS 2 through rosbridge**

3. Differences between ROS2 and ROS1

3.1. Platform

Currently ROS1 only supports running on Linux systems, and is commonly built and used in Ubuntu. ROS2, on the other hand, can be built and used in Ubuntu, Windows, and even on embedded development boards, so it covers a wider range of platforms.

3.2. Language

- C++

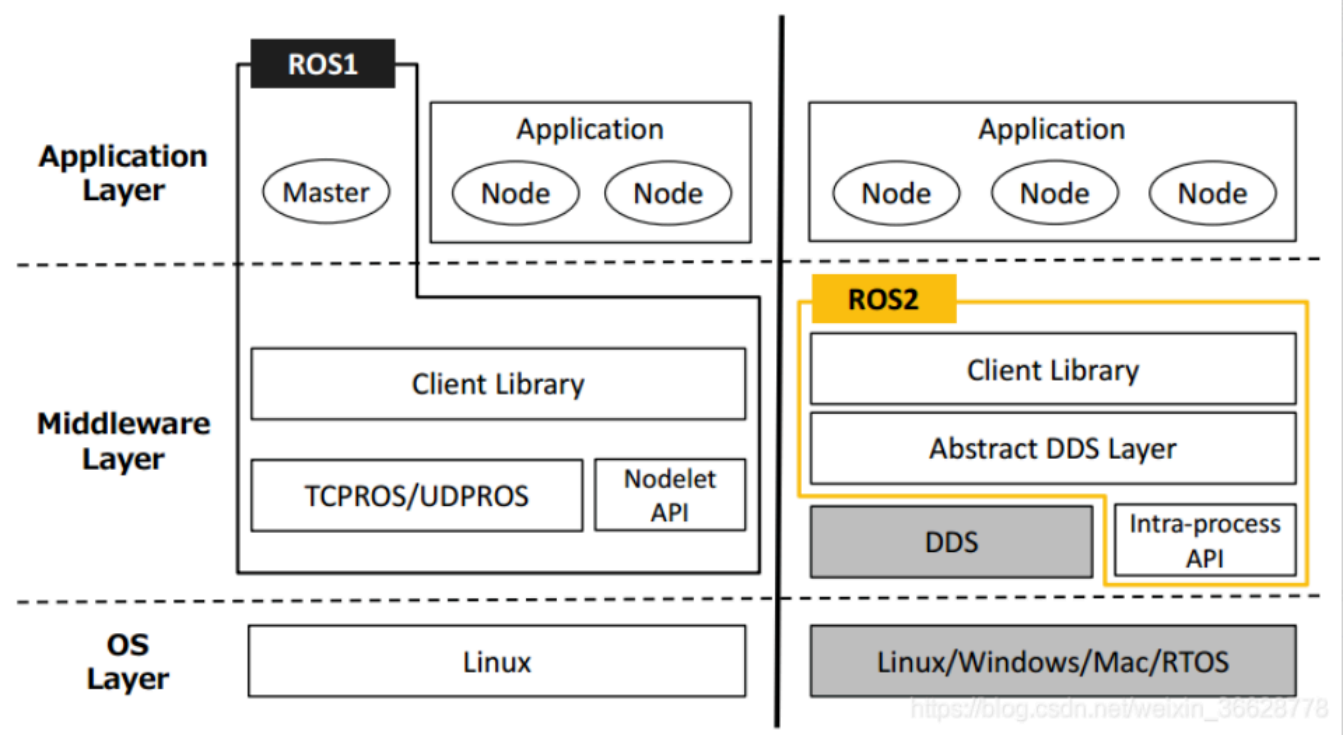
  The core of ROS1 is C++03, while ROS2 makes extensive use of C++11.

- Python

  ROS1 mainly used Python2 in its early days, while ROS2 uses Python3; ROS2 Jazzy on Ubuntu 24.04 uses Python 3.12.

3.3. Middleware

ROS1 needs to start roscore before launching, and this master handles all the communication between nodes, while ROS2 does not have it. Instead, there is only an abstract middleware interface, and data is transferred through this interface. Currently, all implementations of this interface are based on the DDS standard. This allows ROS 2 to provide various high-quality QoS policies, thereby improving communication over different networks.



3.4. Compilation command

The compilation command of ROS1 is catkin_make, while the compilation command of ROS2 is the colcon build command.